AGE 316:
MICROCOMPUTER SYSTEMS

LECTURE 2: HARDWARE & ARCHITECTURE

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Recap: The Brain of the Operation

Recall from Lecture 1: The Microcontroller is a

Computer on a Chip.

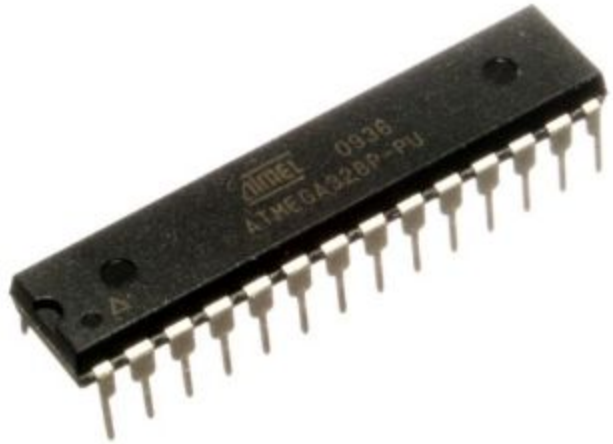
ATmega328P

Manufacturer: Atmel (Microchip)

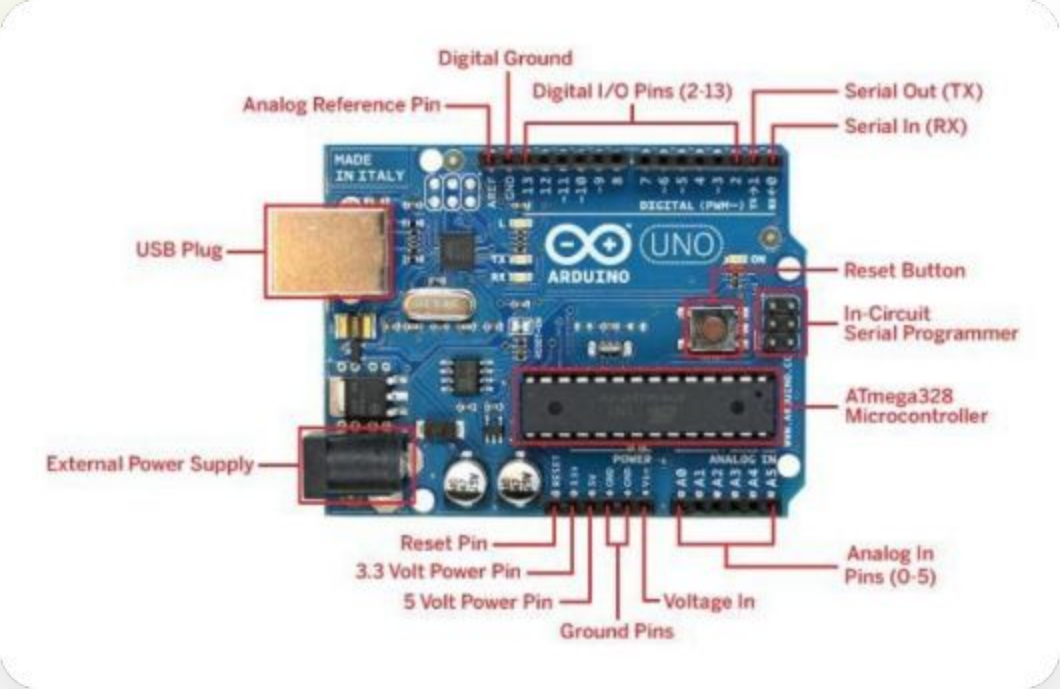
Architecture: AVR 8-bit RISC

Speed: 16 MHz Clock

"This small chip handles all the logic, sensing, and control for your projects."



Dissecting the Arduino Uno



1. **USB Port:** Power & Code upload.
2. **DC Power Jack:** External power (7-12V).
3. **Digital Pins:** On/Off signals & PWM.
4. **Analog Pins:** Reading sensors.
5. **Reset Button:** Restart program.

Power Supply: The Lifeblood

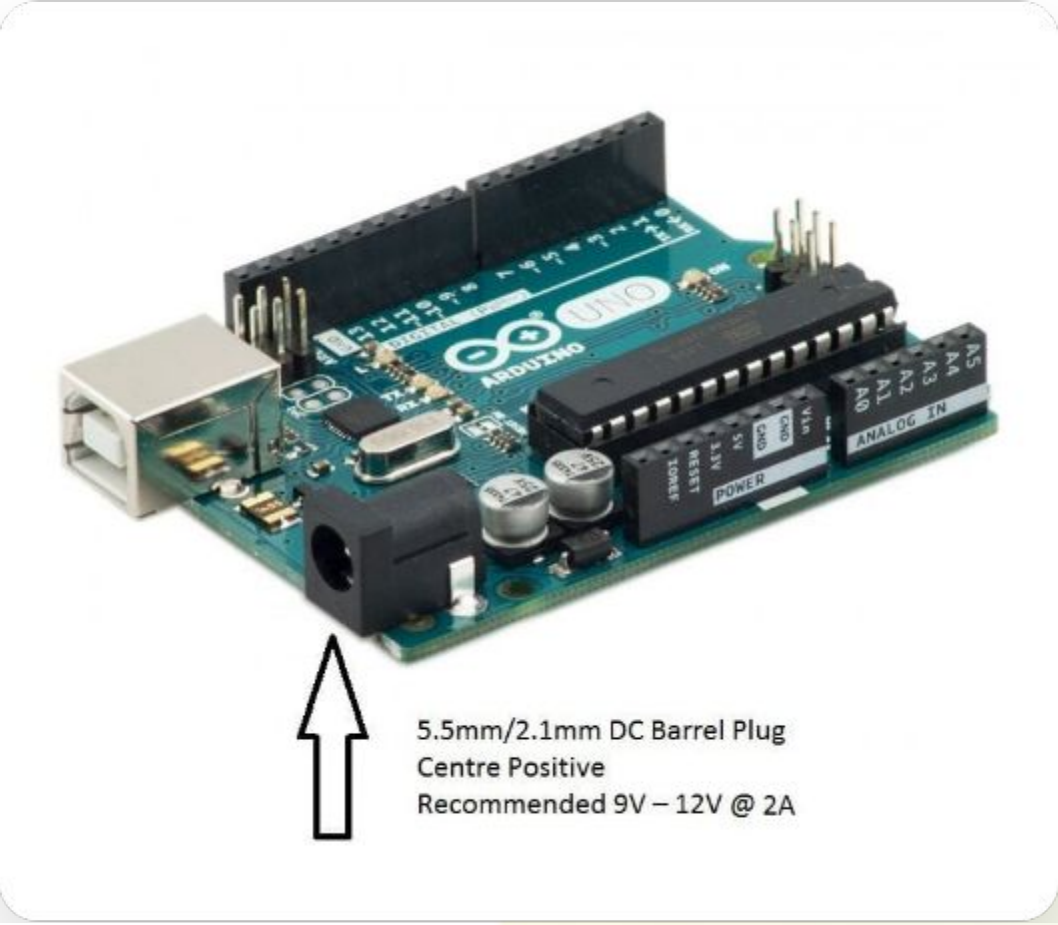
Option A: USB Cable

Supplies regulated **5V**. Best for programming and low-power sensors.

Option B: DC Barrel Jack

Input: **7V - 12V**. Best for standalone projects (batteries/wall adapter). Onboard regulator steps it down.

⚠WARNING:
Do not supply >12V or reverse polarity! You will fry the voltage regulator.



Input / Output (I/O) Pins

Digital I/O

Pins 0 - 13

States: HIGH (5V) or LOW (0V).

Special: PWM (~) on pins 3,5,6,9,10,11 for simulated analog output.

Analog Input

Pins A0 - A5

Function: Reads voltage (0-5V).

Conversion: 10-bit ADC converts voltage to integer (0 - 1023).

Power Pins

Power Header

- **5V:** Regulated output
- **3.3V:** For low voltage sensors
- **GND:** Ground (0V)

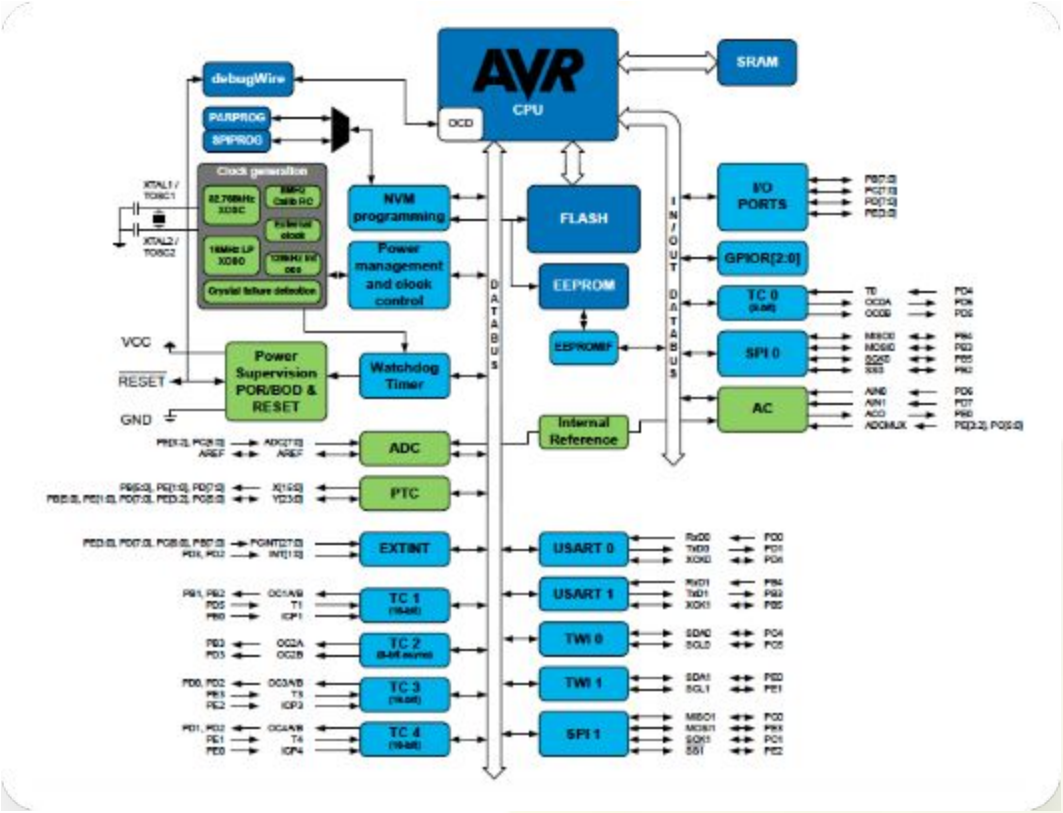
Memory: Inside the Chip

Just like a laptop, the microcontroller has different types of memory for different jobs.

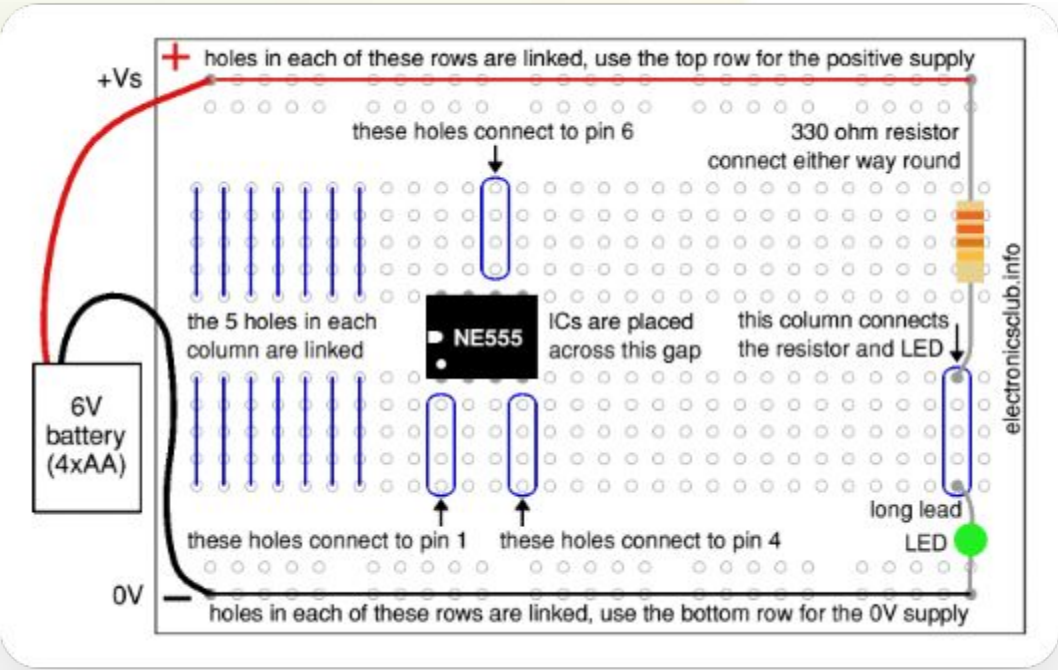
Flash Memory (32 KB)
Your Sketch (Code) lives here. Non-volatile (saved when power off).

SRAM (2 KB)
Variables & Calculations. Volatile (lost when power off).

EEPROM (1 KB)
Long-term storage settings. Non-volatile.



The Breadboard: Circuit Sandbox



How it connects:

Terminal Strips:
Connected **Horizontally** (rows of 5 holes). Used for components.

Power Rails:
Connected **Vertically** (+ and -). Used to distribute power.

Tip: Never put both legs of a component in the same row!

Summary & Next Steps

Today's Key Concepts

- **ATmega328P** is the brain.
- Power via **USB (5V)** or **Jack (7-12V)**.
- **Digital** vs **Analog** pins.
- **Flash** stores code, **SRAM** runs it.

Next Lecture: Software!

We will open the **Arduino IDE** and write our first line of code.



Bring your laptops!

Questions?

"The only stupid question is the one you don't ask."

Dr. Kadeghe Fue

Sokoine University of Agriculture

Image Sources



https://electropeak.com/media/catalog/product/cache/a99a51fafac039a73087ecfaa8ccceba/s/2/s2-0004.jpg

Source: electropeak.com



https://storage.googleapis.com/filo-question-diagrams/6979428_yjapyR_1757351792319062270_image_crop_297ef571-ec6e-4f32-9784-15cbb278a9cc3690835564718814858.jpg

Source: askfilo.com



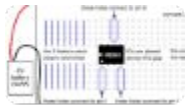
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